Exercise Sheet 7

Selected Probabilistic Algorithms

Probabilistic Machine Learning

Prof. Dr. Michael Granitzer

7.A Mixtures and EM

Exercise 7.A.1: One EM Iteration by Hand

Four points $x = (0, 2, 3, 6)$, two Gaussian components with **fixed** $\sigma_1 = \sigma_2 = 1$, starting from $\pi = (0.5, 0.5)$ and $\mu = (1, 5)$.

1. Show that the responsibility of component 2 is $\gamma_{i2} = 1/\left(1 + e^{12 - 4x_i}\right)$ and evaluate it for the four points.
2. Do one M-step: compute N_1, N_2, the new π and the new μ_1, μ_2.
3. What does k-means (same start) do with the point $x = 3$ in its first assignment step?

Exercise 7.A.2: What EM Still Drops

1. Take a Gaussian mixture with two components and at least two distinct data points. Set $\mu_1 = x_1$ and let $\sigma_1 \to 0$, keeping component 2 fixed and wide. What happens to the likelihood — and why is that not a good model? Why does k-means not have this problem?

2. EM is run for $K = 1, \ldots, 4$. The log-likelihoods are:

K	1	2	3	4
training	−412.3	−371.8	−365.1	−360.4
held-out	−208.9	−186.5	−188.2	−191.7

   Which K do you choose, and why not the one with the best training value?

3. In which sense is EM a *partial* shortcut? Name what it conditions on the data and what it keeps as a point.

7.B Bayesian Regression

Exercise 7.B.1: Bayesian Regression in One Parameter

A price model **through the origin**: $y = wx + \varepsilon, \varepsilon \sim \mathcal{N}(0, 40^2)$, with x the living area in units of 100 m² and y in k€. Prior $w \sim \mathcal{N}(0, 1000^2)$. Data:

x	0.6	0.8	1.0	1.2
y	215	275	355	420

1. Compute the posterior precision $1/s_N^2 = 1/\tau^2 + \sum x_i^2/\sigma^2$, then s_N and $m_N = s_N^2 \sum x_i y_i/\sigma^2$.
2. Give the predictive mean, noise sd, parameter sd and total sd at $x = 1$ (100 m²) and at $x = 6$ (the 600 m² villa).
3. Four villas of 600 m² are added. Recompute the parameter sd and the total sd at $x = 6$ (the posterior mean hardly moves).
4. Instead, a thousand more flats of **100 m²** are added. In this model the parameter sd at 600 m² falls to about 7.6 k€. Why — and why should that make you suspicious, compared with the lecture's model with an intercept?

Exercise 7.B.2: The Two Terms in numpy

Reproduce the lecture's table ($\sigma = 40$ k€, prior $\mathcal{N}(\mathbf{0}, \mathrm{diag}(500^2, 10^2))$). Fill in the three blanks.

```
import numpy as np
x = np.array([40, 44, 48, 53, 57, 61, 65, 69, 74, 78,
              82, 86, 91, 95, 99, 103, 107, 112, 116, 120.])
y = np.array([157, 207, 114, 241, 199, 188, 200, 204, 248, 294,
              299, 314, 255, 272, 384, 379, 430, 415, 357, 443.])
sigma, S0 = 40.0, np.diag([500.0**2, 10.0**2])

def posterior(x, y):
    X = np.c_[np.ones_like(x), x]
    SN = ...                                  # (1) posterior covariance
    mN = ...                                  # (2) posterior mean
    return mN, SN

mN, SN = posterior(x, y)
for q in [80, 120, 300, 600]:
    v = np.array([1.0, q])
    param_var = ...                           # (3) parameter term
    print(q, v @ mN, sigma, np.sqrt(param_var), np.sqrt(param_var + sigma**2))
```

Then add ten villas, `x_v = np.linspace(550, 650, 10).round()` with `y_v = [1900, 1984, 2014, 2056, 2133, 2068, 2133, 2148, 2239, 2325]`, and print the row for 600 m² again.

Exercise 7.B.3: Which Term?

For each situation, say whether the extra uncertainty shows up in the **noise term** σ^2, in the **parameter term** $\mathbf{x}^{*\top}\mathbf{S}_N\mathbf{x}^*$, or in **neither** — and what would reduce it.

1. Two flats with identical area sold a month apart for €310k and €370k.
2. The model is asked about a 600 m^2 villa.
3. The model is asked about a 600 m^2 villa, and villas actually sell at a luxury premium of 5,000 €/m^2.
4. The training data contain only 5 flats instead of 20, all of 40–120 m^2.
5. A lender lends up to the mean minus two sd. Why does the lecture's lender gain about €0.3M of credit on the villa from ten comparable sales, but almost nothing on an 80 m^2 flat from a thousand more flats?